

Lab Work for Week 1: Introduction to Django

Objective

By the end of this lab, students will:

1. Set up the development environment for Django.
 2. Create a new Django project and app.
 3. Understand Django's project structure.
 4. Start the development server and explore the admin interface.
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Step-by-Step Lab Tasks

1. Install Django and Create the Environment

- **Task:** Set up the environment for Django development.
 - Install Python (if not already installed).
 - Create a virtual environment for the project:

```
python -m venv edulearn_env
```
 - Activate the virtual environment:
 - Windows: `edulearn_env\Scripts\activate`
 - macOS/Linux: `source edulearn_env/bin/activate`
 - Install Django:

```
pip install django
```
 - Verify the installation:

```
django --admin --version
```
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2. Create the Django Project

- **Task:** Initialize the project named **EduLearn**.
 - Run the following command:

```
Set -- ExecutionPolicy Unrestricted -- Force  
django --admin startproject EduLearn
```
 - Navigate to the project directory:

```
cd EduLearn
```
 - Understand the structure of the project:

- `manage.py`: A command-line utility for interacting with the project.
 - `EduLearn/`: Contains project settings, URLs, and WSGI/ASGI files.
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3. Start the Development Server

- **Task:** Run the server to ensure the project is set up correctly.
 - Use the following command:

```
python manage.py runserver
```
 - Open a browser and visit `http://127.0.0.1:8000/` to see the Django welcome page.
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4. Create a Django App

- **Task:** Create an app named `courses` for the e-learning platform.
 - Run the following command:

```
python manage.py startapp courses
```
 - Explore the app structure:
 - `apps.py`: App configuration.
 - `models.py`: For defining database models.
 - `views.py`: For handling requests and returning responses.
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5. Register the App

- **Task:** Add the app to the project's settings.
 - Open `EduLearn/settings.py` and add `'courses'` to the `INSTALLED_APPS` list:

```
INSTALLED_APPS = [  
  
    ...,  
    'courses',  
]
```

6. Explore Django's Admin Interface

- **Task:** Access the admin interface.
 - Run the following command to apply default migrations:

python manage.py makemigrations

python manage.py migrate

- Create a superuser:

python manage.py createsuperuser

- Enter a username, email, and password when prompted.

- Start the server again and visit the admin interface at *http://127.0.0.1:8000/admin/*.
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Lab Deliverables

1. A fully functional Django project named EduLearn.
2. An app named courses registered in the project.
3. A superuser account for accessing the admin panel.
4. A screenshot of the Django welcome page and admin interface.